

Every requirement, backed by evidence.

كل مطلب، له دليل.

AI-powered regulatory compliance and application preparation for products entering the Jordanian market.

THE PROBLEM

One product. Several regulators. Scattered requirements.

Before a regulated product reaches the Jordanian market, someone in the importing company has to answer, mostly by hand:

- 1 Which rules apply to this product?
- 2 Which evidence proves each requirement?
- 3 What is missing or inconsistent?
- 4 How do we fill the official application?



\$26.8bn

of merchandise imported into Jordan in 2024

151 pages

of technical standards for telecom devices alone, on top of a 24-article instruction

WHY IT MATTERS FOR JORDAN

Less friction today, a more competitive Jordan tomorrow



The chain is the expected mechanism; MUTABIQ's own effect will be measured in pilots with real importers.

WHY NOW

AI is finally good at document-heavy work, and Jordan is going digital



-40%

+18%

time

quality

for professionals using generative AI on writing and document tasks, in a controlled experiment with 453 participants

Noy & Zhang, *Science*, 2023



JD 3.9bn

projected size of Jordan's digital economy by 2033, up from JD 0.9bn, with all government services to be digitised by 2027

US ITA, Jordan Country Commercial Guide



\$320bn

potential contribution of AI to Middle East GDP by 2030, about 11% of the region's economy

PwC Middle East, 2018

Aligned with

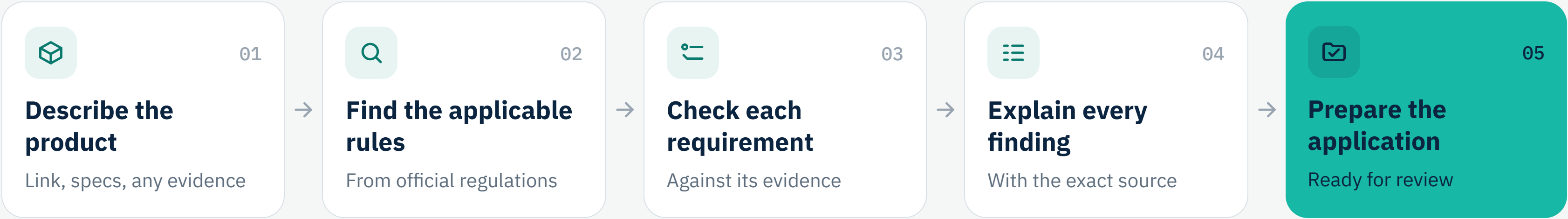
Economic Modernisation Vision 2033

National Digital Transformation Strategy 2021–2025

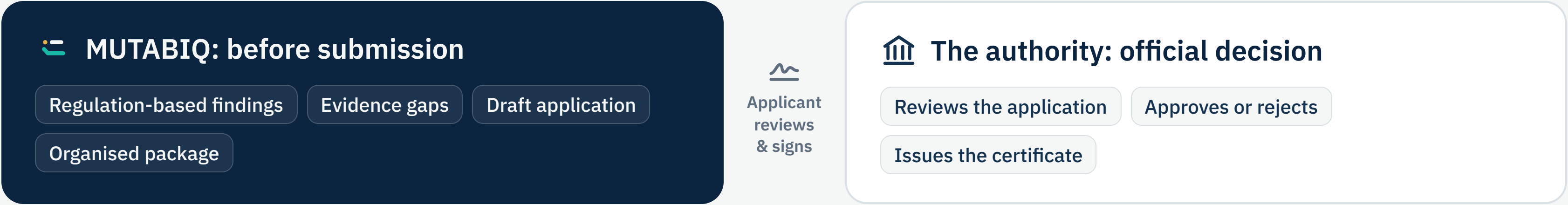
WTO Trade Facilitation Agreement

THE SOLUTION

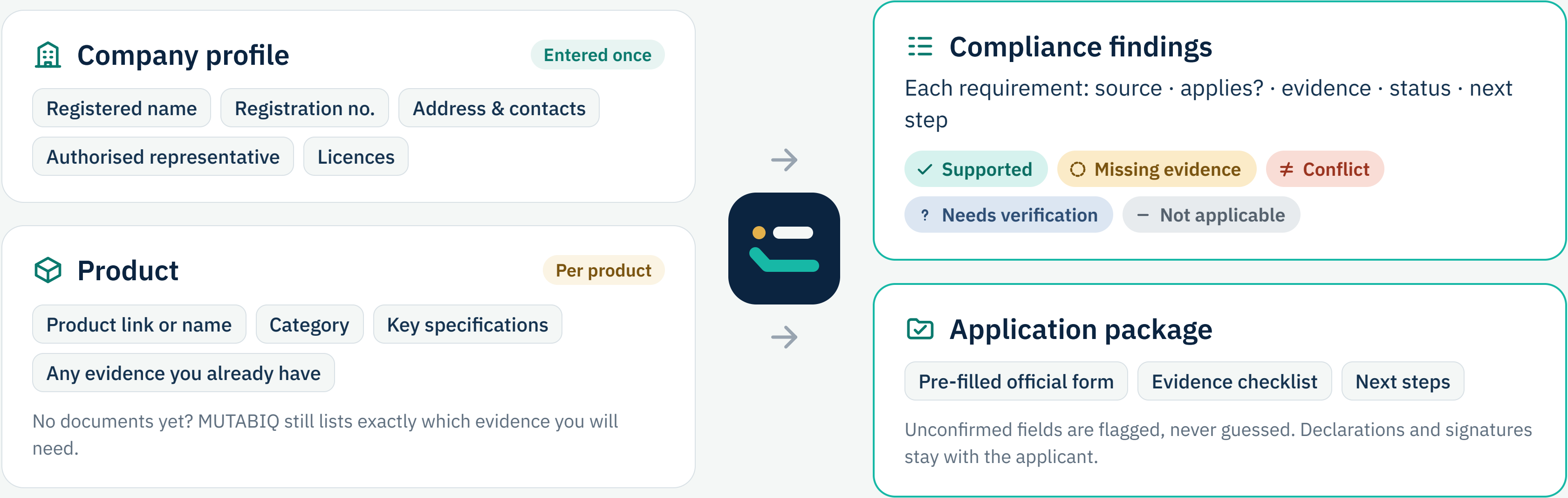
MUTABIQ: a compliance co-pilot for any imported product



📦 A new product category means adding its regulations to the knowledge base. The engine stays the same.



Two inputs, two outputs



Technical trial: one real regulatory case

Trial scope: telecom devices · TRC

The platform is generic. The trial covers one regulatory case, TRC type approval under Instructions No. 2 of 2025. The TRC texts are indexed and searchable live; the devices assessed are sample presets.

DEMO FLOW

1 Enter the inputs0:30

2 Run the assessment0:15

3 Review the findings0:35

4 Find the clause in live search0:15

5 Generate the application draft0:25

New assessment

1 · Inputs2 · Findings3 · Application

Illustrative screen · sample data

COMPANY

Sample Trading Co.

✓ Saved

PRODUCT

Category

Telecom device: smartphone (TRC)

Product link

demo-phone.example/x1

Evidence (optional)

DoC_DEMO-X1.pdfDatasheet.pdf

PRODUCT DATA · SAMPLE PRESET

Brand	DemoTel	Datasheet, p.1	✓
Model	DEMO-X1	DoC	✓
Manufacturer	DemoTel Devices Ltd.	DoC	✓
Frequency range (Tx/Rx)	Bands listed	Datasheet, p.2	✓
Bandwidth	20 MHz	Datasheet, p.2	✓
RF output power (EIRP)	Not found: please enter	–	⦿

Run assessment →

HOW IT WORKS

Evidence retrieval, grounded in official texts

Knowledge base

Built once, re-ingested when a regulation changes



Official source PDFs

Laws, bylaws, TRC instructions, annexes, standards list



Text recovery

Text layer, font decoding or OCR, cross-checked



Split by article & clause

Amendment history kept apart; definitions indexed per term



PostgreSQL + pgvector

bge-m3 embeddings; every version kept by SHA-256

↓ relevant passages retrieved, with citations

Each assessment

Runs per product, in seconds



Product & company inputs

Sample presets in the trial



Planned

AI extraction from DoCs & datasheets



Retrieve the relevant clauses

bge-m3, Arabic & English, article lookup



Planned

LLM explains, from retrieved text only



Deterministic rule checks

TRC requirements written as code



Findings with citations + draft form



Grounded

Results are official text as published, not text written by a model.



Traceable

Every passage carries its article, page, version hash and source URL.



Updatable

A new or amended regulation is re-ingested as a new version, not retrained.

STACK

Python 3.12

PyMuPDF + pdfminer

Tesseract OCR

PostgreSQL 17 + pgvector

BAAI/bge-m3, run locally

Web UI + local JSON API

Trust by design: AI that shows its work



 **No source, no finding**

Every citation names its article, page, source URL and document version.

 **Fixed rules for clear checks**

Presence of a document or matching model numbers is checked by code.

 **"Needs verification", not guesses**

Uncertain values, OCR'd text and unverified legal status are flagged.

 **Missing ≠ non-compliant**

Absent evidence is reported as a gap, not a violation.

 **Human review & signature**

The applicant confirms every field and signs.

 **The authority decides**

MUTABIQ never issues approvals or certificates.

From one trial to a national compliance layer



Every requirement, backed by evidence.

كل متطلب، له دليل.

MUTABIQ prepares the file. The authority makes the decision. Jordanian businesses reach the market with fewer avoidable delays.



APPLICATION READINESS

- Applicable rules found ✓
- Evidence matched to each requirement ✓
- Gaps flagged before submission ✓
- Every finding linked to its source ✓
- Official form pre-filled ✓

Ready for applicant review

Decision: the authority